Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/21/2026
PI Name:	Kolkowitz, Shimon Ph.D.		
Project Title:	Developing New Techniques for Ultra-High-Precision Space-Based Optical Lattice Clock Comparisons		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2022 Physical Sciences NNH22ZDA001N-FP E.6 Fundamental Physics
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No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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We propose a 3-year, ground-based research program dedicated to developing and demonstrating new techniques that will enable ultra-high-precision clock comparisons between space-based optical lattice clocks (OLCs).
Despite rapid progress in OLC performance over the past two decades, there is a huge performance gap between cutting-edge OLCs operating in research laboratories and space-ready atomic clocks. Lab-based OLCs typically occupy multiple optical tables, consume kW-levels of electrical power, require cooling water, and employ large vacuum chambers with bulky pumps. Portable versions currently under development retain much of the complexity and size, weight, and power (SWaP) of their laboratory counterparts. Simply shrinking down and repackaging existing OLC technologies is unlikely to result in space-based OLCs performing at the levels currently demonstrated in research laboratories. In addition, operation of an OLC in microgravity presents both unique challenges and opportunities. New

Task Description:	and innovative approaches and techniques are required. The PI has recently demonstrated a first-of-its-kind “multiplexed" OLC that enables differential clock comparisons between two or more spatially resolved ensembles of strontium atoms within the same vacuum chamber. These differential measurements eliminate the detrimental effects of clock laser noise and common mode environmental fluctuations, pushing the limits of achievable clock stability and atom-atom coherence. Most recently, we have used this apparatus to perform a blinded, precision test of the gravitational redshift at the millimeter to centimeter scale, and to demonstrate relativistic geodesy measurements with millimeter scale height resolution. Over the course of this 3-year ground-based research program, we will harness the unique capabilities of multiplexed OLCs to explore and demonstrate new techniques to substantially reduce the complexity and SWaP requirements of OLCs, and to enable high performance clock comparisons in space. Our efforts will consist of three main thrusts. The first thrust will focus on ultra-high-precision differential clock comparisons over an optical baseline using "noisy" kHz linewidth lasers, without the need for a frequency comb. The second thrust will focus on demonstrating the use of two ensembles in a single clock to double the operational Ramsey free-precession time and thereby significantly enhance the absolute frequency stability of optical lattice clocks. The third thrust will focus on characterizing and demonstrating a new "magic wavelength" that has the potential to substantially reduce the optical power requirements and the number of lasers required to operate a strontium OLC. Taken together, these three advances will represent a major step towards realizing ultra-high-precision optical clock comparisons in space. The realization of high performance space-based OLC networks would result in major technological impacts on NASA missions and goals. For example, the PI has proposed the use of space-based networks of OLCs linked over optical baselines as gravitational wave detectors with unique capabilities. Such networks could also be used to search for several promising dark matter candidates. Clock comparisons between terrestrial optical clocks and space-based clocks will be sensitive to the height of the Earth-based clocks with respect to the geoid at the centimeter scale, complementing NASA gravitational mapping missions such as the Gravity Recovery and Climate Experiment (GRACE), while comparisons with optical clocks placed in eccentric orbits would enable tests of relativity at entirely new levels of precision. OLCs will enable laser ranging and doppler tracking between satellites at new levels of sensitivity for gravitational mapping and deep space navigation. Terrestrial navigation will also benefit from the enhanced robustness of global navigation satellite systems provided by space-based OLCs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Ultra-high-precision comparisons across space-based clock networks have applications to gravitational wave detection, laser ranging, doppler tracking of spacecraft, gravitational mapping, redefinition of spectroscopic imaging (SI) units, and terrestrial navigation.
Task Progress:	The project team currently consists of Principal Investigator (PI) Kolkowitz, postdoctoral researchers Dr. Shuo Ma and Dr. Dhruva Ganapathy, graduate student Anke Stoeltzel, undergraduate researcher Jenny Chen, and post-baccalaureate researcher Bennett Christensen. Graduate student Jack Dolde successfully defended his thesis and received his PhD from UC Berkeley during the reporting period. Undergraduate researcher Jenny Chen also received her bachelor's degree in physics from UC Berkeley during this reporting period, and will start a PhD program in physics at CU Boulder/JILA in the fall of 2026. Post-baccalaureate researcher Bennett Christensen received his bachelor's degree in physics from UC Berkeley in May 2025 and has continued to contribute to the project as a post-bacc; he will start a PhD program in physics at Colorado State University in the fall of 2026. Over the course of the second year of performance we have continued to make significant progress towards the goals of this grant. Two manuscripts that were originally submitted to peer-reviewed journals during the first year of performance, and that were posted to the physics arXiv at that time, have now both been published. Our experimental demonstration that hyperfine-state-resolved erasure conversion can be used to dramatically extend the achievable atomic coherence time on the strontium-87 clock transition has been published in PRX Quantum. [Ed. Note: See Bibliography.] In this work we demonstrated atomic coherence times in excess of 150 seconds, more than an order of magnitude longer than what had previously been demonstrated in any optical atomic clock. The extension of coherence times in differential comparisons is particularly relevant for applications of space-based optical clocks of interest to NASA, such as dark matter searches and gravitational wave detection. In the process we also developed and demonstrated new techniques for performing and analyzing differential comparisons utilizing dynamical decoupling sequences, which are critical for performing dark matter searches and testing sequences for gravitational wave detection. Our companion measurement of the natural radiative lifetime of the upper strontium-87 clock state, in which we definitively observed radiative decay on the strontium-87 clock transition for the first time and measured the value for the natural radiative lifetime of this transition to be 167 (+79) (-40) seconds, has been published in Physical Review A. We have also made major experimental progress towards characterizing and demonstrating a new "magic wavelength" for strontium optical lattice clocks at 497 nm. Operation of strontium optical lattice clocks at a 497 nm magic wavelength has the potential to substantially reduce the optical power requirements for these clocks, which is particularly relevant for operation of an optical lattice clock in space. Following the installation of a custom 497 nm Vertical-External-Cavity Surface-Emitting Laser (VECSEL) during the first year of performance, we have now successfully trapped strontium-87 atoms in a 497 nm optical lattice. We have experimentally confirmed the substantially enhanced atomic polarizability of strontium at this wavelength, observing significantly deeper traps for the same optical power compared to the conventional 813 nm magic wavelength, as expected. We have also observed clean clock transition spectra in the 497 nm lattice, confirming that high quality coherent interrogation of the strontium-87 clock transition is possible at this wavelength. In addition, we have performed detailed measurements of the photon scattering rate out of the upper clock state as a function of 497 nm trap depth. The scattering rate is a critical figure of merit for the practical utility of a 497 nm magic-wavelength lattice for optical lattice clock operation in space, and to our knowledge has not been previously reported. During the reporting period an independent measurement of the magic wavelength of the strontium clock transition was performed for the bosonic isotope strontium-88 at 497.4364(4) nm by Kestler et al. and published in Physical Review Applied. Building on our 497 nm lattice trapping, clock spectra, and scattering rate measurements, we are currently working on a precision experimental determination of the 497 nm magic wavelength in strontium-87. In parallel with our ongoing experimental work, postdoctoral researcher Dr. Dhruva Ganapathy has led a detailed

	theoretical and protocol-development study of the use of space-based optical clock networks for gravitational wave astronomy. This work develops dedicated one-way Doppler measurement protocols, analyzes the dominant noise sources, and performs preliminary parameter estimation on simulated gravitational wave signals from binary compact objects observed by a heliocentric space-based clock network. It is directly relevant to the broader goals of this grant, and motivates the differential clock comparisons with "noisy" local oscillators that we will pursue during the third and final year of performance. The manuscript is in the final stages of preparation, and we expect to post it to the physics arXiv and submit it to Physical Review D within the next several weeks. Finally, as part of broader engagement with the community on emerging frequency reference technologies that are relevant to NASA's fundamental physics goals, the PI consulted on a collaborative effort led by D. Budker and V. Malinovsky at the U.S. Army Research Laboratory which experimentally demonstrated a temperature-compensated composite nitrogen-vacancy-center diamond clock, and which has been accepted for publication in Physical Review Applied. Although solid-state nitrogen-vacancy-center clocks are operationally distinct from the optical lattice clocks at the center of this grant, the underlying scientific questions about how to reduce environmental sensitivity, suppress dominant systematic shifts, and engineer robust, compact frequency references are directly relevant to the broader goals of this grant. This NASA grant is acknowledged in the publication. During the third and final year of the grant we plan to complete a precision experimental determination of the 497 nm magic wavelength in strontium-87, to demonstrate ultra-high-precision differential clock comparisons with kHz-linewidth local oscillators by intentionally degrading the spectral purity of our clock laser, and to continue to extend the multi-ensemble and erasure conversion techniques that we developed during the first two years of performance towards prototype protocols for dark matter searches and gravitational wave detection.
Bibliography Type:	Description: (Last Updated: 07/07/2026)
Articles in Other Journals or Periodicals	Lourette S, Jarmola A, Chathanathil J, Acosta VM, Birdwell AG, Blumler P, Budker D, Carrasco SC, Ivanov TG, Kolkowitz S, Malinovsky VS. "Towards a temperature-insensitive composite diamond clock." arXiv preprint server. Posted January 1, 2026. https://arxiv.org/abs/2601.00157 , Jan-2026
Articles in Peer-reviewed Journals	Lourette S, Jarmola A, Chathanathil J, Acosta VM, Birdwell AG, Blümler P, Budker D, Carrasco SC, Ivanov TG, Kolkowitz S, Malinovsky VS. "Toward a temperature-insensitive composite diamond clock." Phys Rev Applied 2026 Jun 11;25(6):064046. https://doi.org/10.1103/z2sl-6gcc , Jun-2026
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Articles in Peer-reviewed Journals	Dolde J, Ganapathy D, Zheng X, Ma S, Beloy K, Kolkowitz S. "Direct measurement of the 3P0 clock state natural lifetime in 87Sr." Physical Review A. 2025 Aug 28;112(2):023121. https://doi.org/10.1103/f6pt-flnt , Aug-2025